

Conjugacy in $SL_2(F_3)$ II

Arham Lodha

November 14, 2023

We continue aiming for $24 = 1 + 1 + 6 + 4 + 4 + 4 + 4$ and recall we have $S \rightarrow PS = A_4 \subset PG = S_4$ = permutation of the 4 one dim subspaces of PV , $V = F_3^2$. What are the possible characteristic polys for a matrix in S ? $X^2 + 1$, $X^2 - 2X + 1 = (X - 1)^2$, $X^2 + 2X + 1 = (X + 1)^2$. (note $2 = -1$ in F_3).

Problem 10a. *Show that a one dimensional subspace of V preserved by s is the same thing as a fixed point of the permutation $q(s)$ in $PS = A_4$.*

Proof. Let $s \in S$ s.t s preserves a one dimensional subspace of V , l . Thus $l \subseteq V_\lambda(s)$. Then $q(s) = sZ$ preserves the eigenspace $V_\lambda(s)$. If $\dim V_\lambda(s) = 2$, then s scales all vectors and thus $s \in Z$ and thus $q(s) = Z$ and the corresponding permutation is the identity and l is a fixed point in the permutation. If $\dim V_\lambda(s) = 1$, then $V_\lambda(s) = l$ and thus if sZ preserves the eigenspace $sZ(l) = l$ thus l is fixed in the corresponding permutation. $\square$

Problem 10b. *Suppose s is not in the center (not I or $-I$). Show that $p(s)$ is a 3-cycle iff s has an eigenvalue (in F_3) which is iff its characteristic polynomial factors.*

Proof. Suppose $s \notin Z(S)$.

($\implies$) Suppose s has a eigenvalue λ in F_3 , then λ is a root of its characteristic polynomial thus its characteristic polynomial factors.

($\impliedby$) Suppose the characteristic polynomial of s factors. Thus there exists a root of the characteristic polynomial in $F_3 \implies s$ has a eigenvalue in F_3 by definition.

($\implies$) $p(s)$ is a 3 cycle. Then there is a 1 dimensional subspace which is a fixed point in $p(s)$. Thus s preserves the one dimensional subspace thus the one dimensional subspace is in a eigenspace of s .

($\impliedby$) Suppose s has a eigenvalue, λ . Then $V_\lambda(S) \neq \emptyset$. Since $s \notin Z(S)$. Then $\dim V_\lambda(S) = 1$ since $s \notin F^*$. By the previous part we know that if s preserves a 1 dimensional subspace the subspace is a fixed point in $p(s)$. The only permutations in A_4 which have a fixed point are I and any three cycle. Since $s \notin Z$, $p(s) \neq Z = I$ so $p(s)$ must be a 3 cycle. $\square$

Problem 11a. *Show that if s and $-s$ are conjugate, then $Tr(s) = 0$.*

Proof. The trace of any two conjugate matrices are equal, because 2 conjugate matrices have the same eigenvalues. $s \sim -s \implies Tr(s) = Tr(-s) = -Tr(s)$ because trace is linear. Thus $Tr(s) = 0$. $\square$

Problem 11b. *Show that if s and $-s$ are conjugate, then $p(s)$ has no fixed point, so has to be in the conjugacy class 3.*

Proof. If $s \sim -s \implies \text{Tr}(s) = 0$ (by problem 11a). $P_s(x) = x^2 + \text{Tr}(s)x + \det(s) = x^2 + \det(s)$. The characteristic polynomial is not factorable. Thus s has no eigenvalue in F^3 thus there is no fixed point in $p(s)$. The conjugacy class of 3 corresponds to the conjugacy class of the (i j)(k l) which have no fixed points. All other non 1 conjugacy classes correspond to 3 cycles which have fixed points. Thus it has to be conjugacy class of 3. $\square$

Problem 11c. Conclude, from the theory in the previous homework that the "inverse image of $1 + 3 + 4 + 4$ " gives $1 + 1 + 6 + 4 + 4 + 4 + 4$.

Proof. The 1 corresponds to I whose inverse image is I and $-I$ which are both in the center. So it becomes $1 + 1$. Since elements in the conjugacy class of 3 elements are conjugate to their negatives, it doubles and turns into 3 and 3. The elements in the conjugacy class of 4 elements have characteristic polynomial of $(x - 1)^2$ or $(x + 1)^2$. For the first conjugacy class with the characteristic polynomial of $(x - 1)^2$ a representative matrix is $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. $A \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \begin{bmatrix} a & a+b \\ c & c+d \end{bmatrix} = \begin{bmatrix} -a-c & -b-d \\ -c & -d \end{bmatrix}$ thus $c = -c$ and $c = 0$ and A is not invertible and thus a matrix is not conjugate to its negative. Thus the 4 breaks up. $-A$ corresponds to the other conjugacy class of 4 and since $A \not\sim -A$ both 4s break up. Thus $4 + 4 = 4 + 4 + 4 + 4$. Thus the inverse image of $1 + 3 + 4 + 4$ is $1 + 1 + 6 + 4 + 4 + 4 + 4$. $\square$

Problem 12. Now write down representatives for each of the 7 conjugacy classes

Proof. Representative of conjugacy class of 1: I .

Representative of conjugacy class of 1: $-I$.

Representative of conjugacy class of 3: $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Representative of conjugacy class of 4: $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Representative of conjugacy class of 4: $\begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix}$

Representative of conjugacy class of 4: $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

Representative of conjugacy class of 4: $\begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$

$\square$

Problem 13. Here we did $S = SL_2$. Do "the same thing" for $G = GL_2$. E.g. consider the 3-cycles, which are an "8" in the class equation for S_4 . Explain why, from what we already did (and our theory from the previous two homeworks) why necessarily the inverse image in G has to be $8 + 8$. Also explain why the inverse image of the $4 = 2 + 2$ conj class (product of two disjoint 2-cycles) has to be 6. and the inverse image of 1 has to be $1 + 1$. How about 2-cycles? explain why $p(g)$ is a 2-cycle for non-central s iff s is diagonalizable, with eigenvalues 1, -1. Show that s and $-s$ are conjugate, conclude the "6" in the class equation for S_4 has inverse image 12 in the class equation for G . Finish the analysis.

Proof. The class equation for S_4 is $1 + 3 + 6 + 6 + 8$.

In $G = GL_2(F_3)$ all the characteristic equations are:

1. $X^2 + 1$
2. $X^2 - 2X + 1 = (X - 1)^2$
3. $X^2 + 2X + 1 = (X + 1)^2$
4. $X^2 - 1 = (X - 1)(X + 1)$
5. $X^2 + 1X - 1$
6. $X^2 - 1X - 1$

For the conjugacy class of size 8. We know its entirely contained as a set in A_4 as the conjugacy class of $4 + 4$. We know that $SL_2(F_3)$ is the preimage of $PSL_2(F_3)$ of the function $GL_2(F_3) \rightarrow PGL_2(F_3)$. We know that the preimage of the $4 + 4$ is $4 + 4 + 4 + 4$. The possibilities for the preimage conjugacy class of size 8 in S_4 are Conjugacy class of 16 or conjugacy classes $8 + 8$. We know that in the map from $GL_2(F_3) \rightarrow SL_2(F_3)$ the classes need to break up into $4 + 4 + 4 + 4$, this is not possible if they are 16 as if 16 breaks up it breaks up into $8 + 8$ (past homework). So $8 + 8$ is the only possibility of the preimage of the Conjugacy class of size 8.

The permutations with matrices with characteristic polynomial 4 are permutations with 2 fixed points so a single 2 cycle and they correspond to one of the 6s. So a matrix like $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ are Conjugate because they have the same eigenvalues. Thus the 6 doesn't split up and becomes 12.

Permutations in the other conjugacy classes of 6 elements correspond to the matrices with characteristic polynomials of 5 and 6, matrices with no eigenvalues and thus no fixed points and they are not in A_4 . The matrices corresponding to 4 cycles. One such matrix is $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ which is not conjugate to its negative after checking. Thus 6 splits up to be $6 + 6$.

Since the conjugacy class of 3 stays together in A_4 it must also stay together in S_4 , because we know its elements are conjugate to its negative in $SL_2(F_3)$.

Thus the class equation is $1 + 1 + 6 + 6 + 12 + 8 + 8$. □